

Learning how to prove Workshop Week 4

Permutations



Cycles

1. The **order of a permutation** $\sigma \in S_n$ is the smallest positive integer m such that $\sigma^m = \varepsilon$.
 - (a) If σ is an r -cycle, prove that then σ has order r . In other words, the order of a cycle equals its length.
 - (b) If $\sigma = \gamma_1 \dots \gamma_s$ is a product of disjoint r_i -cycles γ_i then σ has order $m = \text{lcm}(r_1, \dots, r_s)$.
 - (c) Use (b) to find the order of the following permutations:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 5 & 7 & 9 & 14 & 10 & 11 & 12 & 8 & 3 & 13 & 2 & 6 & 4 & 1 \end{pmatrix}$$

$$\tau = (1 \ 2)(3 \ 4 \ 5 \ 6)$$